Chapter 19

The Red Planet

Mars regolith as a chem+ system: iron oxidation states, contamination pathways, and the engineering chemistry of surface processing.

Contents

19.0. Core Idea

Mars is red because the surface is dominated by oxidised iron-bearing minerals. In chem+ language, the visible planet is basically a planetary-scale contamination layer:

$$\left[\![\,\text{Mars Regolith}\,]\!]_{s}^{\text{Fe/O/Si/Al/Mg/Ca/Cl/S}}\right]$$

with the red identity carried mainly by:

$$\left[\![\,\mathrm{Fe_2O_3}\,]\!]_{s}^{\text{Fe}-\text{O}-\text{Fe}}\right]$$

$$\left[\![\,\text{FeOOH}\,]\!]_{s}^{\text{Fe}-\text{O}-\text{H}}\right]$$

$$\left[\![\,\mathrm{Fe_3O_4}\,]\!]_{s}^{\mathrm{Fe^{2+}/Fe^{3+}}}\right]$$

So the surface colour is not "Mars is red," like a children's book written by a committee. It is:

$$\left[\![\,\text{Fe}\,]\!]_{\text{reduced/rock, }s}\right] \rightarrow \left[\left[\!\!\left[\,\mathrm{Fe^{3+}}-\text{O}\,\right]\!\!\right]_{s}^{\text{oxidised dust}}\right]$$

The planet is red because iron has been converted into oxidised surface-state solids.

19.1. Red Planet Identity Vector

Define the regolith identity vector:

$$\vec{R}_{\text{Mars}} = \begin{bmatrix} \mathrm{SiO_2} \\ \mathrm{Fe_2O_3} \\ \mathrm{Al_2O_3} \\ \text{MgO} \\ \text{CaO} \\ \mathrm{SO_4^{2-}} \\ \mathrm{ClO_4^-} \\ \mathrm{H_2O_{ads}} \end{bmatrix}$$

In compressed chem+ notation:

$$\left[\![\,\text{Regolith}\,]\!]_{s}^{\text{silicate/oxide/sulfate/perchlorate}}\right]$$

Expanded:

$$\begin{aligned} [\![\,\text{Regolith}\,]\!]_{\text{Mars}} = {} & [\![\,\text{Si}-\text{O}-\text{Si}\,]\!]_{s} + [\![\,\text{Fe}-\text{O}-\text{Fe}\,]\!]_{s} + [\![\,\text{Al}-\text{O}-\text{Si}\,]\!]_{s} \\ & + [\![\,\text{Mg}-\text{O}\,]\!]_{s} + [\![\,\text{Ca}-\mathrm{SO_4}\,]\!]_{s} + \left[\!\!\left[\,\mathrm{ClO_4^-}\,\right]\!\!\right]_{salt} + [\![\,\mathrm{H_2O}\,]\!]_{ads} \end{aligned}$$

The iron oxide piece is the red-state carrier:

$$\left[\llbracket \mathrm{Fe{-}O{-}Fe} \rrbracket_{s}^{\text{red chromophore}}\right]$$

Yes, the planet's branding department is rust. Space has no shame.

19.2. Red-State Formation Pathway

A simplified oxidation path:

$$\left[\llbracket \mathrm{Fe} \rrbracket_{\text{metal/silicate, } s}\right] + \llbracket \mathrm{O_2} \rrbracket_{atm} + \llbracket \mathrm{H_2O} \rrbracket_{ads} \rightarrow \llbracket \mathrm{FeOOH} \rrbracket_{s}$$

Then dehydration / ageing:

$$2\llbracket \mathrm{FeOOH} \rrbracket_{s} \rightarrow \llbracket \mathrm{Fe_2O_3} \rrbracket_{s} + \left[\llbracket \mathrm{H_2O} \rrbracket_{g/ads}\right]$$

Chem+ flow:

$$\left[\llbracket \mathrm{Fe} \rrbracket^{\text{reduced}}\right]_{\text{rock}} \rightarrow \left[\llbracket \mathrm{Fe{-}OH} \rrbracket_{\text{surface}}^{\text{weathered}}\right] \rightarrow \left[\llbracket \mathrm{Fe{-}O{-}Fe} \rrbracket_{\text{dust}}^{\text{oxidised}}\right]$$

Main state transition:

$$\mathrm{Fe^0/Fe^{2+}} \rightarrow \mathrm{Fe^{3+}}$$

Bond transition:

$$\llbracket \mathrm{Fe{-}Si/O} \rrbracket_{\text{rock}} \rightarrow \llbracket \mathrm{Fe{-}OH} \rrbracket_{\text{weathered}} \rightarrow \llbracket \mathrm{Fe{-}O{-}Fe} \rrbracket_{\text{oxide}}$$

19.3. The Red Planet Contamination Lesson

The earlier acid-leach contamination pathway becomes important because Mars regolith contains iron oxides everywhere. The danger is not merely that iron exists. The danger is this conversion:

$$\left[\llbracket \mathrm{Fe} \rrbracket_{s}^{\text{immobile oxide}}\right] \rightarrow \left[\llbracket \mathrm{Fe} \rrbracket_{aq}^{\text{mobile ferric}}\right]$$

On Mars, this matters for:

⟦ water extraction ⟧	⟦ oxygen production ⟧	⟦ regolith processing ⟧
⟦ construction binders ⟧	⟦ metal extraction ⟧	⟦ sample purity ⟧

because any acidic or reactive processing can drag iron into the liquid phase and ruin separation.

Humanity goes to Mars and immediately has to ask: "How do we not dissolve the entire red problem into our product stream?" Charming species. Very focused.

19.4. Revised Mars Contamination Pathway

Using the earlier pathway, the full acid-dissolution sequence for a surface iron oxide is:

$$\begin{aligned}
&\Big[\big[\,Fe_2O_3\,\big]_s^{Fe^{3+}-O^{2-}-Fe^{3+}}\Big] \\
&\xrightarrow{+\,H^+} \Big[\big[\,Fe{-}OH{-}Fe\,\big]_{surface}^{protonated}\Big] \\
&\xrightarrow{+\,H^+} \Big[\Big[\,Fe{-}OH_2^+\,\Big]_{surface}^{leaving}\Big] \\
&\xrightarrow{cleavage} \Big[\Big[\,Fe^{3+}\,\Big]_{aq}^{mobile}\Big] \\
&\xrightarrow{+\,H_2O} \Big[\Big[\,Fe(H_2O)_6^{3+}\,\Big]_{aq}^{aquo}\Big] \\
&\xrightarrow{+\,SO_4^{2-}} \Big[\Big[\,FeSO_4^+\,\Big]_{aq}^{inner/outer\ sphere}\Big] \\
&\xrightarrow{cluster} \Big[\big[\,Fe{-}\mu SO_4{-}Fe\,\big]_{aq}^{bridged}\Big]
\end{aligned}$$

Mars version, combining multiple dissolved iron species:

$$\begin{aligned}
&\Big[\big[\,\text{Mars dust}\,\big]_s^{Fe_2O_3/FeOOH}\Big] + \Big[\big[\,\text{acidic brine}\,\big]_{aq}^{H^+/SO_4^{2-}/Cl^-}\Big] \\
&\rightarrow \Big[\Big[\,Fe^{3+}\,\Big]_{aq}^{mobile}\Big] + \Big[\Big[\,FeSO_4^+\,\Big]_{aq}\Big] + \Big[\Big[\,FeCl^{2+}\,\Big]_{aq}\Big] + \Big[\Big[\,FeOH^{2+}\,\Big]_{aq}\Big]
\end{aligned}$$

So the Mars red-state problem becomes:

red dust $\rightarrow$ dissolved ferric contamination $\rightarrow$ dirty brine chemistry

19.5. Chem+ Bond-Pattern Library for Mars Iron

A. Hematite-like Oxide

$$\Big[\big[\,Fe_2O_3\,\big]_s^{Fe^{3+}-O^{2-}-Fe^{3+}}\Big]$$

Bond motif:

$$\mathrm{Fe{-}O{-}Fe}$$

State:

$$\Big[\,[\,\mathrm{Fe}\,]_{+3,\ \mathrm{oxide}}\Big]$$

B. Hydroxide / Oxyhydroxide Surface

$$\Big[\,[\,\mathrm{FeOOH}\,]_{s}^{\mathrm{Fe{-}O{-}H}}\Big]$$

Bond motif:

$$\mathrm{Fe{-}OH}$$

Surface hydrated form:

$$\Big[\,[\,\mathrm{Fe{-}OH}\,]_{\mathrm{dust}}^{\mathrm{surface}}\Big]$$

C. Magnetite-like Mixed Iron State

$$\Big[\,[\,\mathrm{Fe_3O_4}\,]_{s}^{\mathrm{Fe^{2+}/Fe^{3+}}}\Big]$$

State vector:

$$\vec{F}_{\mathrm{mag}} = \begin{bmatrix} \mathrm{Fe^{2+}} \\ \mathrm{Fe^{3+}} \\ \mathrm{O^{2-}} \end{bmatrix}$$

Mixed redox identity:

$$\Big[\,[\,\mathrm{Fe_3O_4}\,]_{s}^{\mathrm{mixed\ valence}}\Big]$$

This is important because mixed-valence iron can shift under chemical processing:

$$\mathrm{Fe^{2+}} \rightleftharpoons \mathrm{Fe^{3+}} + e^-$$

D. Ferric Aquo Contaminant

$$\Big[\Big[\,\mathrm{Fe^{3+}}\,\Big]_{aq}\Big] + 6\,[\![\,\mathrm{H_2O}\,]\!]_{l} \rightarrow \Big[\Big[\,\mathrm{Fe(H_2O)_6^{3+}}\,\Big]_{aq}\Big]$$

Chem+:

$$\Big[\,[\,\mathrm{Fe}\,]^{+3}\Big]_{aq} \cdot \Big[\,[\,\mathrm{OH_2}\,]_{6}^{\sigma\ \mathrm{donor}}\Big]$$

Bond motif:

$$\mathrm{Fe{-}OH_2}$$

E. Ferric Sulfate Contaminant

$$\left[\left[\mathrm{Fe}^{3+}\right]_{aq}\right] + \left[\left[\mathrm{SO}_4^{2-}\right]_{aq}\right] \rightleftharpoons \left[\left[\mathrm{FeSO}_4^{+}\right]_{aq}\right]$$

Chem+:

$$\left[\left[\mathrm{Fe{-}O{-}SO_3}\right]_{aq}^{\text{sulfate bound}}\right]$$

Bond motif:

$$\mathrm{Fe{-}O{-}S}$$

F. Bridged Ferric Sulfate Cluster

$$2\left[\left[\mathrm{Fe}^{3+}\right]_{aq}\right] + \left[\left[\mathrm{SO}_4^{2-}\right]_{aq}\right] \rightarrow \left[\left[\mathrm{Fe{-}O{-}SO_2{-}O{-}Fe}\right]_{4+,\,aq}\right]$$

Chem+:

$$\left[\left[\mathrm{Fe{-}}\mu\mathrm{SO_4{-}Fe}\right]_{aq}^{\text{bridged}}\right]$$

Bond motif:

$$\mathrm{Fe{-}O{-}S{-}O{-}Fe}$$

This is where the dissolved iron stops being a clean ion and becomes a coordination mess wearing a fake moustache.

19.6. Mars Processing Comparison

Acidic Route

$$\left[\left[\text{Mars Regolith}\right]_{s}^{\mathrm{Fe_2O_3/SiO_2/Al_2O_3}}\right] + \left[\left[\mathrm{H}^{+}\right]_{aq}\right]$$
$$\rightarrow \left[\left[\mathrm{Fe}^{3+}\right]_{aq}\right] + \left[\left[\mathrm{Al}^{3+}\right]_{aq}\right] + \left[\left[\mathrm{Mg}^{2+}\right]_{aq}\right] + \left[\left[\mathrm{Ca}^{2+}\right]_{aq}\right] + \left[\left[\mathrm{SiO_2}\right]_{\text{residue}}\right]$$

Problem:

$$\left[\left[\mathrm{Fe_2O_3}\right]_{s}\right] \rightarrow \left[\left[\mathrm{Fe}^{3+}\right]_{aq}\right]$$

Acid makes extraction stronger but dirtier.

Basic Route

$$[\![\,\text{Mars Regolith}\,]\!]_s + \left[\left[\,\text{OH}^-\,\right]_{aq}\right] \to \left[\left[\,\text{silicate/aluminate}\,\right]_{aq}\right] + \left[\left[\,\text{Fe}_2\text{O}_3\,\right]_{s/\text{residue}}\right]$$

Ideal selectivity:

$$[\![\,\text{Fe}_2\text{O}_3\,]\!]_s \not\to \left[\left[\,\text{Fe}^{3+}\,\right]_{aq}\right]$$

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basic leach keeps more iron locked as solid oxide

The base route may be less aggressive, but it is cleaner for keeping red iron out of solution. Naturally, the slower path is the more civilised one. Annoying but typical.

19.7. Regolith State Matrix

Define the Mars regolith state matrix:

$$\mathcal{R}_{\text{Mars}} = \begin{bmatrix} \text{Fe}_2\text{O}_3 & \text{FeOOH} & \text{Fe}_3\text{O}_4 \\ \text{SiO}_2 & \text{Al}_2\text{O}_3 & \text{MgO} \\ \text{CaSO}_4 & \text{NaCl/MgCl}_2 & \text{ClO}_4^- \end{bmatrix}$$

Rows:

$$\begin{bmatrix} \text{redox / iron} \\ \text{silicate structure} \\ \text{salts / reactive contaminants} \end{bmatrix}$$

Columns:

$$\begin{bmatrix} \text{oxide} & \text{hydrated / surface} & \text{reactive / mobile} \end{bmatrix}$$

A processing operation acts on this matrix:

$$\mathcal{P} : \mathcal{R}_{\text{Mars}} \to \mathcal{R}'_{\text{Mars}}$$

Four canonical operators:

$$\mathcal{P}_{\text{acid}} : [\![\,\text{Fe}-\text{O}-\text{Fe}\,]\!]_s \to \left[\left[\,\text{Fe}^{3+}\,\right]_{aq}\right]$$

$$\mathcal{P}_{\text{base}} : [\![\,\text{Si}-\text{O}-\text{Al}\,]\!]_s \to \left[\left[\,\text{silicate/aluminate}\,\right]_{aq}\right]$$

$$\mathcal{P}_{\text{thermal}} : [\![\,\text{H}_2\text{O}\,]_{ads}]_s \to \left[\left[\,\text{H}_2\text{O}\,\right]_g\right]$$

$$\mathcal{P}_{\text{redox}} : \left[\left[\,\text{Fe}^{3+}\,\right]\right]_s + 3e^- \to \left[\left[\,\text{Fe}^0\,\right]_{s/l}\right]$$

19.8. Red Planet Extraction Decision

The main engineering question:

Do we want iron mobile, fixed, reduced, or separated?

Path 1 — Preserve iron as solid contamination

$$[\![Fe_2O_3]\!]_s \rightarrow [\![Fe_2O_3]\!]_{residue}$$

Useful when extracting water, salts, or light elements.

Path 2 — Dissolve iron intentionally

$$[\![Fe_2O_3]\!]_s + 6 \left[\!\left[H^+ \right]\!\right]_{aq} \rightarrow 2 \left[\!\left[Fe^{3+} \right]\!\right]_{aq} + 3 [\![H_2O]\!]_l$$

Useful for chemical analysis, but bad for clean product streams.

Path 3 — Reduce iron to metal

$$[\![Fe_2O_3]\!]_s + 3 \left[\!\left[H_2 \right]\!\right]_g \rightarrow 2 [\![Fe]\!]_s + 3 \left[\!\left[H_2O \right]\!\right]_g$$

Chem+:

$$\left[\!\left[Fe^{3+}{-}O \right]\!\right]_s \rightarrow \left[\!\left[Fe^0 \right]\!\right]_s$$

Useful for metallurgy and construction feedstock.

Path 4 — Convert oxide oxygen into useful oxygen

$$[\![MO]\!]_s \rightarrow \left[\!\left[M \right]\!\right]_{s/l} + \left[\!\left[O_2 \right]\!\right]_g$$

General oxygen extraction:

$$\left[\!\left[\text{oxide} \right]\!\right]_{regolith} \rightarrow [\![\text{metal/silicate}]\!] + [\![O_2]\!]$$

On Mars, oxygen extraction is not just life support. It is propellant, combustion, welding, processing, and not dying stupidly in a very expensive desert.

19.9. The Red Planet as a Chem+ System

Final chapter identity:

$$\left[\!\left[\text{Mars} \right]^{\text{red}}_{\text{planet}}\right] = [\![\text{silicate body}]_{\text{rock}}] + \left[\!\left[\text{Fe}^{3+}{-}\text{O} \right]_{\text{dust}}\right] + [\![\text{salts/perchlorates}]_{\text{surface}}] + [\![\text{CO}_2]_{\text{atm}}] + \left[\!\left[\text{H}_2\text{O} \right]_{\text{ice/ads}}\right]$$

The useful abstraction:

$$[\![\text{Mars surface}]\!] = [\![\text{structure}]\!] + [\![\text{oxidation}]\!] + [\![\text{salt}]\!] + [\![\text{dust}]\!] + [\![\text{volatile}]\!]$$

Or in one clean chem+ statement:

$$[\![\text{Red Planet}]\!] = \left[\!\left[\text{Fe}{-}\text{O}{-}\text{Fe} \right]^{\text{colour/state}}_{\text{dust}}\right] + \left[\!\left[\text{Si}{-}\text{O}{-}\text{Si} \right]^{\text{structure}}_{\text{rock}}\right] + \left[\!\left[\text{SO}_4^{2-}/\text{ClO}_4^{-} \right]^{\text{reactive salts}}_{\text{regolith}}\right] + \left[\!\left[\text{CO}_2/\text{H}_2\text{O} \right]^{\text{volatile}}_{\text{atm/ice}}\right]$$